STRATHMORE UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

MACHINE LEARNING-BASED TOOL FOR EARLY DETECTION OF AT-RISK STUDENTS IN HIGHER EDUCATION: CASE OF STRATHMORE UNIVERSITY

BY

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DECLARATION

Student's Declaration

I declare that this thesis is my original work and has not been presented to any other institution for examination or any other award. Where other sources of information have been used, they have been properly acknowledged.

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Supervisor's Declaration

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Abstract

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List of Algorithms

Table of Abbreviations

Abbreviation	Description
AMS	Academic Management System
ANN	Artificial Neural Network
AUC	Area Under the Curve
BI	Business Intelligence
CBET	Competency-Based Education and Training
CESA	Continental Education Strategy for Africa (2016–2025)
DT	Decision Tree
EAB	Education Advisory Board
EDM	Educational Data Mining
EWS	Early Warning System
F1	F1-Score (Harmonic Mean of Precision and Recall)
GPA	Grade Point Average
GPS	Graduation and Progression Success
GSU	Georgia State University
HEI	Higher Education Institution
I-E-O	Input-Environment-Output (Astin's Model)
KCSE	Kenya Certificate of Secondary Education
KIPPRA	Kenya Institute for Public Policy Research and Analysis
LIME	Local Interpretable Model-agnostic Explanations
LMS	Learning Management System
LR	Logistic Regression
LSTM	Long Short-Term Memory
ML	Machine Learning

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Abbreviation	Description
MLP	Multilayer Perceptron
NBT	National Benchmark Test
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SHAP	SHapley Additive exPlanations
SIM	Student Integration Model
SMOTE	Synthetic Minority Oversampling Technique
SVM	Support Vector Machine
TAN	Tree Augmented Naïve Bayes
UCT	University of Cape Town
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Definition of Terms

Term	Description
Academic Integration	The degree to which a student engages with the academic system of the institution, measured through grade performance (GPA) relative to institutional standards and normative alignment with academic community values (Tinto, 1975).
Academic Management System (AMS)	The primary institutional database for official student records, including demographic information, enrolment status, GPA trajectories, and academic history.
At-Risk Student	A student whose data patterns statistically indicate a high probability of dropout, course failure, or programme delay prior to the actual event.
Attendance Threshold	The minimum attendance rate required for examination eligibility, set at 67% at Strathmore University. Breach renders a student exam-ineligible regardless of academic performance.
Black Box Problem	The inability of complex ML models to provide human-interpretable explanations for their predictions, creating an institutional trust deficit that limits advisor adoption.
Class Imbalance	A condition in student risk datasets where at-risk students constitute a minority of the total population, leading to models biased toward the majority class if uncorrected.

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Term	Description
Course Failure Risk	The predicted likelihood of a student failing a course or becoming exam-ineligible due to attendance falling below the mandatory 67% threshold at Strathmore University.
Data Pipeline	An automated, integrated system that extracts, merges, cleans, and validates raw data from AMS, LMS, and attendance platforms before feeding it into machine learning models.
Dropout Risk	The predicted probability of a student discontinuing enrolment before completing their degree requirements.
Early Warning System (EWS)	An integrated tool using data and predictive modelling to trigger timely alerts for students requiring academic intervention before failure thresholds are breached.
Educational Data Mining (EDM)	An interdisciplinary field applying computational and statistical techniques to large-scale educational datasets to identify patterns enhancing teaching and institutional decision-making.
Engagement Metrics	Quantitative measures of student participation: LMS login frequency, assignment timeliness, resource access counts, and attendance rate trajectories.
Ensemble Learning	A machine learning technique combining multiple algorithms—Random Forest, Gradient Boosting, XGBoost—to produce more accurate and stable predictions than any single model alone.
Explainability	The capacity of a predictive model to provide human-interpretable reasons for its predictions, typically implemented through SHAP values.

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Term	Description
F1-Score	The harmonic mean of Precision and Recall, providing a single balanced metric accounting for false alarms and missed detections.
Fairness	The requirement that model predictions do not systematically disadvantage students on the basis of protected or socioeconomic characteristics.
Feature Engineering	The process of transforming raw institutional data into optimised ML input variables, including attendance breach flags, GPA floor proximity scores, and temporal engagement patterns.
GPA Floor	The minimum GPA (2.0 at Strathmore University) for good academic standing; breach triggers academic probation.
Historical Cohort	A defined student group whose complete academic trajectories are known and documented, used as training data for predictive models.
Human-in-the-Loop	A deployment principle requiring ML models to support rather than replace advisor judgment; the model supplies evidence while the advisor decides.
Hyperparameter Tuning	Systematic optimisation of algorithm configuration settings to ensure reported performance reflects a properly calibrated model.
Intervention	A proactive support action (mentoring, tutoring referral, advisor outreach) triggered by risk status to prevent failure or withdrawal.

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Term	Description
Learning Management System (LMS)	A digital platform delivering course content and tracking engagement via login frequency, resource access, and assignment submissions.
Model Accuracy	The proportion of correct predictions (at-risk and not-at-risk) relative to total observations.
Overfitting	A modelling failure where an algorithm memorises training data noise rather than generalisable patterns, inflating performance metrics.
Precision	The proportion of students predicted as at-risk who are genuinely at-risk; minimises misallocation of scarce counselling resources.
Predictive Analytics	A branch of data analytics applying statistical modelling to historical data to forecast future student outcomes.
Programme Delay Risk	The probability that a student will exceed standard programme duration due to probation, course repetition, or accumulated failed semesters.
Reactive Support Paradigm	A support model where students are identified only after confirmed failure indicators, by which point recovery is often impossible.
Recall (Sensitivity)	The proportion of genuinely at-risk students correctly identified; the theoretically primary metric in student support contexts.
Risk Stratification	Classification of students into risk categories (low/medium/high) to prioritise advisor outreach toward the most vulnerable.
ROC-AUC	A threshold-independent discrimination metric; 1.0 = perfect, 0.5 = chance-level performance.

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Term	Description
Role-Based Dashboard	A user interface presenting SHAP-based risk predictions tailored to each actor: university administrator, school administrator, or academic mentor.
SHAP (SHapley Additive exPlanations)	An explainable AI technique decomposing each prediction into individual feature contributions, enabling student-level risk explanation.
SMOTE	Synthetic Minority Oversampling Technique; generates synthetic at-risk instances during training to correct class imbalance.
Social Integration	The degree of meaningful peer and faculty relationships, operationalised via LMS engagement and attendance proxies (Tinto, 1975).
Student Attrition	Withdrawal from higher education before degree completion, causing financial loss to students and revenue loss to institutions.
Student Integration Model (SIM)	Tinto's (1975) framework positing persistence depends on academic and social integration; dropout is a longitudinal process, not a discrete event.
Supervised Learning	An ML paradigm training models on labelled historical data to predict class membership of new unlabelled instances.
XGBoost	Extreme Gradient Boosting; a high-performance ensemble algorithm identified as top-performing across multiple African HEI dropout prediction studies.

Chapter 1

INTRODUCTION

1.1 Background to the Study

Student retention is a persistent structural challenge in higher education worldwide. Undergraduate enrolment in U.S. degree-granting institutions declined by 13% between 2012 and 2022, with 23.3% of students leaving annually; six-year graduation rates stand at 68% for private nonprofits, 63% for public universities, and only 29% for for-profit institutions (Irwin et al., 2024b). A systematic review of 125 peer-reviewed studies concluded that, despite decades of institutional intervention, dropout remains a globally significant and structurally unresolved problem (Gonçalves et al., 2024).

Two complementary frameworks explain the mechanisms driving this attrition. Tinto's Student Integration Model (Tinto, 1975) holds that persistence depends on a student's ongoing academic and social integration—framing dropout as a longitudinal process in which declining GPA and reduced engagement signal deteriorating belonging rather than isolated academic failure. Astin's Input-Environment-Output (I-E-O) Model (Astin, 1993) provides the variable architecture, categorising student inputs, institutional environmental conditions, and measurable outcomes, and confirming that factors such as advisor interaction, LMS engagement, and attendance are independently associated with retention (Xiao et al., 2020). Together, these frameworks define the feature selection and outcome structure of this study.

Machine learning has transformed how institutions can respond to these dynamics. Predictive algorithms can now identify at-risk students weeks before conventional warning signs appear, achieving accuracies of 88–92% (Ahmed, 2024; Gathani et al., 2026). A

systematic review of 72 Educational Data Mining (EDM) studies confirmed that ensemble methods—particularly XGBoost and Random Forest—consistently outperform classical classifiers, while SHAP-based explainability has emerged as a critical prerequisite for institutional adoption (Aldowah et al., 2026; Kwakye, 2025). Georgia State University's GPS Advising system exemplifies the institutional impact achievable: tracking over 800 risk factors for 40,000 students daily raised its graduation rate by 7.8 percentage points and eliminated achievement gaps across demographic groups (**gsu2024_**).

These benefits have yet to reach most African institutions. Mohammed V University in Morocco achieved an AUC-ROC of 0.993 using an open-source XGBoost–SHAP pipeline (Bettahi et al., 2025), and Wollo University in Ethiopia demonstrated that ensemble classifiers reach 96% accuracy on large-scale African institutional data (Esmael, 2024)—yet both studies, alongside South African work at the University of Johannesburg (Taodzera et al., 2016) and the University of Cape Town (Hassen, 2020), share a consistent gap: the absence of student-level SHAP explainability that would make predictions actionable for advisors. In Kenya specifically, the earliest published study—at Machakos University College—identified fee payment difficulty as the dominant attrition predictor, positioning financial exclusion as the primary structural driver of dropout in Kenyan higher education (J. K. Ngemu et al., 2015). Most African universities, however, still rely on end-of-semester transcripts as their primary diagnostic tool.

Strathmore University operates within this landscape while enforcing a 67% minimum attendance threshold for examination eligibility and a 2.0 GPA floor for academic standing—thresholds that make late identification especially costly. The institution generates daily student data through its Academic Management System (AMS), Learning Management System (LMS), and attendance tracking platform, yet these systems operate as isolated repositories with no integrated analytics capability. Students are identified for intervention only after breaching these thresholds, by which point the window for effective recovery has often closed. This study addresses that gap by developing a machine learning-based predictive analytics tool—integrating all three data streams,

predicting dropout, course failure, and programme delay simultaneously, and embedding SHAP explainability throughout—to transition Strathmore from reactive monitoring to proactive, evidence-driven student support.

1.2 Statement of the Problem

Currently, Strathmore University possesses the digital infrastructure necessary for proactive student support—an Academic Management System (AMS), a Learning Management System (LMS), and an attendance tracking platform—yet these systems operate as isolated administrative repositories with no integrated analytics capability. As a result, the institution cannot transform its own institutional data into early warnings for at-risk students. Students are identified for intervention only after breaching the mandatory 67% attendance threshold for examination eligibility or the 2.0 GPA floor for academic standing—by which point the window for effective recovery has often closed. The outcomes that follow are both predictable and preventable: students drop out of programmes they could have completed, fail courses that timely referrals could have salvaged, and accumulate delays that extend their studies by entire semesters—each representing a personal and institutional loss that earlier detection would have averted (**gsu2024_**; Bettahi et al., 2025; Esmael, 2024).

Three interconnected capability gaps sustain this reactive paradigm. First, existing systems cannot detect deteriorating engagement trajectories—declining LMS login frequency, late assignment submissions, and falling attendance—weeks before they manifest as failed grades or breached thresholds, despite research confirming that such time-based signals carry greater predictive power than static GPA values alone (Dirin & Saballe, 2022a; Morris et al., 2005a). Second, no student-level explainability exists: advisors cannot identify which specific factors are driving a student's deterioration and therefore cannot conduct evidence-based intervention conversations—a well-documented barrier to institutional adoption of predictive analytics (Kwakye, 2025; Zhang & Tao, 2021). Third, the binary persist-or-dropout framing of prior tools fails to

capture the full spectrum of academic difficulty; course failure risk and programme delay risk are distinct, actionable outcomes that demand simultaneous, policy-calibrated prediction alongside dropout risk (Bettahi et al., 2025).

This study addresses these gaps by developing, implementing, and evaluating an integrated machine learning-based predictive analytics tool tailored to Strathmore University's academic policies, data infrastructure, and resource constraints. By unifying the AMS, LMS, and attendance data streams into a single risk stratification pipeline, training open-source ensemble models on the 2021–2024 historical student cohort, and embedding SHAP-based explainability throughout, the platform simultaneously predicts dropout risk, course failure risk, and programme delay risk—each calibrated to the institution's 67% attendance threshold and 2.0 GPA floor. The goal is to transition Strathmore from a reactive support model, in which advisors act on crisis, to a proactive one, in which limited counselling resources are directed by evidence to the students who need them most.

1.3 Research Objectives

1.3.1 Main Objective

To design, develop, and evaluate a machine learning-based predictive analytics tool for early identification of at-risk undergraduate students at Strathmore University.

1.3.2 Specific Objectives

1. To design a Machine Learning predictive tool for university administrators, school administrators, and academic mentors that present actionable risk predictions with supporting intervention guidance.

2. To evaluate model accuracy and predictive performance using historical student cohort data (2021–2024) and standard machine learning evaluation metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC.

3. To assess the platform's usability and perceived institutional impact through structured evaluation with academic advisors, school administrators, and mentors, with specific attention to intervention feasibility and long-term sustainability.

1.4 Research Questions

1. How can risk predictions be effectively presented to different stakeholder groups—university administrators, school administrators, and academic mentors—to support informed decision-making?
2. To what extent does training predictive models on historical student cohort data (2021–2024) with known outcomes produce valid and generalisable predictions for current enrolled students?
3. How do academic advisors, school administrators, and mentors perceive the platform's usability, intervention feasibility, and potential for long-term institutional adoption?

1.5 Justification

1.5.1 Operational Necessity

This research is justified by the urgent need to bridge the gap between data collection and data utilisation at Strathmore University. Currently, the institution's student support is reactive—interventions occur only after a student has already failed a course, breached the 67% attendance threshold, or submitted a withdrawal notice. The EDM literature is unambiguous that this reactive paradigm is both preventable and institutionally costly: Georgia State University's GPS Advising system demonstrated that proactive, data-driven intervention can raise graduation rates by 7.8 percentage points and generate over 90,000 annual adviser-student contacts (Georgia State University, 2024a). Strathmore possesses the necessary digital infrastructure—AMS, LMS, and attendance

tracking—but lacks the integrated ML architecture to operationalise it. There is therefore an urgent need for a system that transforms underutilised data into actionable early warnings without increasing the administrative burden on faculty.

1.5.2 Addressing the Explainability Gap

This study is additionally justified by its focus on explainability as a design prerequisite rather than an optional enhancement. A systematic analysis of existing African EDM studies reveals a consistent pattern: high-performing models at Machakos University College, Wollo University, the University of Johannesburg, and the University of Cape Town all lack SHAP-based or equivalent explainability layers, which prevents their outputs from being translated into student-specific intervention recommendations (Esmael, 2024; Hassen, 2020; J. K. Ngemu et al., 2015; Taodzera et al., 2016). The practical consequence is that advisors cannot determine which specific factors triggered a risk classification for an individual student— undermining the evidence-based advising that proactive support requires. By embedding SHAP (SHapley Additive exPlanations) throughout the platform's prediction pipeline, this study operationalises the finding that explainable predictions provide intuitive decision support and guidance for education administrators (Zhang & Tao, 2021), and demonstrates that SHAP-based explainability is not a luxury but a prerequisite for AI adoption in resource-constrained environments (Kwakye, 2025).

1.5.3 Policy and Economic Pressures

This study responds directly to the Kenya Ministry of Education's Sessional Paper No. 1 of 2024, which cites inadequate technology use as a core bottleneck in higher education (Kenya Institute for Public Policy Research and Analysis, 2024). In a climate of rising enrolments and limited government funding—where capitation covers only 57% of students and public universities have accumulated Ksh 62 billion in unpaid bills (Daily Nation, 2024; Kenya Institute for Public Policy Research and Analysis, 2024)—the status quo of universal, intuition-based mentoring is no longer sustainable. The Bet-

tahi et al. (2025) open-source pipeline at Mohammed V University establishes that a high-performance, explainable Early Warning System can be deployed entirely using open-source tools (Python, scikit-learn, XGBoost, SHAP) without commercial licensing costs—directly resolving the affordability constraint that renders the Georgia State GPS model non-transferable to the African context. This research is therefore justified as both an economic necessity—ensuring that finite counselling resources are directed to where they are most needed—and as a replicable policy instrument for comparable East African institutions.

1.6 Scope and Delimitations of the Study

1.6.1 Scope

This research focuses on the development and evaluation of a predictive analytics platform for approximately 5,000 undergraduate students across all five schools at Strathmore University, Kenya. The technical scope involves integrating data from the Academic Management System (AMS), Learning Management System (LMS), and attendance tracking records into a unified analytics pipeline. The study develops three prediction models—targeting dropout risk, course failure risk, and program delay risk—using ensemble algorithms including Random Forest, Gradient Boosting, and XGBoost, trained on historical student cohort data from 2021–2024 with known academic outcomes. Model predictions are accompanied by SHAP-based explainability outputs to ensure advisors and administrators can understand and act on the reasons behind each risk classification (Lundberg & Lee, 2017). The resulting platform includes role-based dashboards for university administrators, school administrators, and academic mentors, with model performance measured through standard machine learning evaluation metrics (Accuracy, Precision, Recall, F1-score, and ROC-AUC) and platform effectiveness assessed through structured stakeholder feedback.

1.6.2 Delimitations

The research boundaries are defined by the following deliberate constraints. Regarding data factors, the analysis is restricted to academic risk factors measurable through institutional systems, such as attendance rates, GPA, LMS engagement, and course grades. External variables such as socioeconomic background, mental health status, and family circumstances are excluded, as they are not systematically captured in Strathmore's existing data infrastructure and their collection would raise ethical concerns beyond the current scope. In terms of student population, the study is limited to undergraduate students across Strathmore University's five schools; postgraduate and continuing education cohorts are excluded due to their unique assessment patterns.

The system capabilities are delimited to identifying risk levels and suggesting general intervention categories, such as academic counselling or tutoring referrals, rather than providing course-specific remediation. Within the algorithmic scope, the study employs ensemble tree-based methods like Random Forest and XGBoost (Chen & Guestrin, 2016) rather than deep learning architectures. This decision is justified by the dataset size of approximately 5,000 students and the critical need for feature interpretability. Furthermore, the technical platform is developed as a web-based application, with mobile-native development remaining outside the scope. Financial domains, including fee payment risk and scholarship management, are also excluded to focus purely on academic and engagement indicators. Finally, the implementation phase concludes with a functional prototype and structured evaluation; full-scale production deployment and long-term longitudinal impact assessment are reserved for future research.

1.7 Significance

1.7.1 Theoretical and Methodological Significance

This study advances Educational Data Mining (EDM) by operationalising Tinto's Student Integration Model and Astin's I-E-O Model within an African institutional context—

mapping Tinto's academic integration construct onto GPA trajectories and attendance breach flags calibrated to Strathmore's 67% threshold, and Astin's environmental variables onto LMS engagement metrics and the 2.0 GPA floor (Astin & Antonio, 2012; Bento Júnior et al., 2025; Tinto, 1975). It further extends prior work by simultaneously predicting dropout, course failure, and programme delay risk—a multi-outcome architecture no prior African EDM study has attempted (Bettahi et al., 2025). Methodologically, the study establishes SHAP-based explainability as a prerequisite—not an optional enhancement—for predictive model adoption in resource-constrained settings (Kwakye, 2025), and provides a rigorous benchmark through ensemble methods, stratified cross-validation, SMOTE oversampling, and hyperparameter tuning (Aldowah et al., 2026; Bello et al., 2025; Esmael, 2024).

1.7.2 Practical Significance

For *institutions*, the study delivers an open-source blueprint for transitioning from reactive to proactive student support, directly resolving the proprietary and cost barriers that make the Georgia State GPS Advising model non-transferable to African universities (Bettahi et al., 2025; Wan, 2019a).

For *educators*, role-based SHAP dashboards shift advisor decision-making from intuition to evidence—enabling reason-led intervention conversations grounded in each student's specific risk drivers rather than crisis-stage metrics.

For *students*, the platform transforms the institution's punitive policy thresholds—the 67% attendance rule and 2.0 GPA floor—into a supportive safety net, enabling academic recovery before those thresholds are irreversibly breached and preserving the conditions for integration and persistence that Tinto's model identifies as central to retention (Tinto, 1993a).

1.7.3 National and Policy Significance

At the national level, this research provides a replicable model for Kenya's transition to Competency-Based Education and Training (CBET) and for the Ministry of Education's broader technology-utilisation agenda (Kenya Institute for Public Policy Research and Analysis, 2024). By demonstrating that a mid-sized African university can deploy high-performance, explainable predictive analytics using exclusively open-source tools—extending the approach validated at Mohammed V University (Bettahi et al., 2025)—the study offers a concrete, cost-effective policy instrument for comparable East African institutions. It further responds to the structural finding from the Machakos University College study that financial exclusion is the dominant attrition driver in Kenyan higher education (J. K. Ngemu et al., 2015), showing that institutional data already collected can generate early warnings precise enough to support targeted counselling and fee-assistance referrals without additional data collection.

1.8 Summary

This chapter established the context, rationale, and direction of the study. Beginning at the global level, it demonstrated that student dropout remains a critical challenge in higher education and that machine learning-based predictive analytics has proven capable of identifying at-risk students weeks before conventional indicators surface—yet these benefits remain largely inaccessible to number of African universities.

The logic of disparate systems forces institutions like Strathmore into a reactive support paradigm where students are identified only after breaching the 67% attendance threshold or 2.0 GPA floor. This gap, situated within Kenya's national call for greater technology utilization in education, defines the problem this study addresses.

The study responds by developing an ML-based predictive analytics platform tailored to Strathmore's policies and resource context. The platform predicts dropout, course failure, and programme delay using ensemble algorithms trained on the 2021–2024

historical cohort. Its contributions span theory, practice, and policy: advancing retention frameworks in African contexts, providing a sustainable and replicable institutional tool, and demonstrating a cost-effective model for comparable universities across the region. Chapter Two now reviews the literature underpinning these contributions, examining theoretical frameworks, documented system implementations with their successes and limitations, and the precise knowledge gaps this study is designed to fill.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

This chapter examines the theoretical and empirical foundations underpinning the development of a machine learning-based early warning system for at-risk students. It begins by situating student attrition within a global context, establishing retention as a persistent structural challenge across higher education systems. It then reviews two dominant theoretical frameworks — Tinto's Student Integration Model and Astin's Input-Environment-Output Model — which together provide the conceptual architecture for feature selection and outcome definition. Secondly, it surveys the field of Educational Data Mining, critically analyzing existing ML-based predictive systems across the United States, Africa, and Kenya as far as regional level is concerned. Finally, it identified methodological gaps this study will be directly addressing.

2.2 Global Perspectives on Student Attrition

Student retention is a global imperative linked directly to both institutional sustainability and the fulfillment of higher education's social mission. According to Irwin et al. (2024a), undergraduate enrollment in degree-granting institutions in the United States decreased by 13% between 2012 and 2022, compounded by a persistent annual attrition rate of 23.3%. Six-year graduation rates reveal substantial structural disparities: 68% at private nonprofit institutions, 63% at public universities, and only 29% at private for-profit institutions. Despite a modest recent recovery—the 2024 National Student Clearinghouse report recorded a national first-year retention rate of 68%, described

as the highest in the past decade (National Student Clearinghouse Research Center, 2024)—the structural problem of attrition remains unresolved, with large proportions of students continuing to exit before degree completion.

These figures carry dual consequences. At the institutional level, each departing student represents lost tuition revenue, wasted administrative investment, and diminished institutional capacity to fulfill its educational mission. At the individual level, students who withdraw often exit the system bearing financial debt without the credential necessary for socioeconomic mobility. A systematic review of 125 peer-reviewed studies on undergraduate retention published between 2014 and 2022 concluded that, despite decades of research and institutional intervention, student dropout remains a persistent and structurally significant problem across higher education globally, with research in the field remaining fragmented and without a unified evidence base for effective remediation (Gonçalves et al., 2024). The following table summarizes graduation rate variation by institution type:

Table 2.1: Six-Year Graduation Rates by Institution Type (Irwin et al., 2024a)

Institution Type	Six-Year Graduation Rate
Private Nonprofit Institutions	68%
Public Universities	63%
Private For-Profit Institutions	29%

Beyond the United States, the problem is not geographically bounded. Widening access policies across the Global South have expanded enrollment without proportional improvements in completion, meaning that participation gains mask a deepening completion deficit (Gonçalves et al., 2024). In sub-Saharan Africa and Kenya specifically, this deficit is compounded by resource constraints, infrastructure gaps, and the financial pressures, making the attrition problem both more acute and less resourced than in high-income institutional contexts. These global statistics therefore establish the urgency of the problem and demonstrate that student attrition is not a peripheral concern but a

structural threat to the viability of higher education institutions worldwide—providing the macro-level context within which Kenya's institutional challenges, and this study's intervention, must be understood.

2.3 Theoretical Frameworks for Student Retention

This section is anchored in two complementary theoretical frameworks that together provide the conceptual architecture for understanding and predicting student attrition. Tinto's Student Integration Model explains the longitudinal mechanism through which diminishing academic and social integration drives departure, while Astin's I-E-O Model provides the variable architecture — categorizing student inputs, institutional environmental conditions, and measurable outcomes — that a predictive system must capture. Both frameworks were developed in Western institutional contexts but carry constructs directly mappable to Strathmore University's data infrastructure.

2.3.1 Tinto's Student Integration Model

The dominant theoretical anchor for student retention research remains Vincent Tinto's Student Integration Model (SIM). First, it was proposed by Tinto and Cullen in 1973 and formally published as a peer-reviewed synthesis in 1975. This model draws its conceptual foundations from Émile Durkheim's theory of suicide— specifically the proposition that a deficit of social cohesion precipitates withdrawal from the community (Tinto, 1975). Tinto's scholarly influence has been exceptional: his works have been cited in more than 700 studies across multiple disciplines, establishing the SIM as the most widely adopted theoretical lens in higher education retention research (Bento Júnior et al., 2025).

At its core, the SIM conceptualizes the university as a microcosm comprising two interdependent systems—the academic and the social—and posits that a student's persistence is governed by their degree of integration into both (Tinto, 1975; Tinto, 1993b). Critically, Tinto frames dropout not as a single discrete event but as a *longitudinal pro-*

cess: the student's commitment to persist is continuously reconfigured through ongoing interactions. This interactional character includes the cumulative weight of how individual attributes, academic performance, and social relationships interact and accumulate over time (Nicoletti, 2019; Tinto, 1993b).

2.3.1.1 The Longitudinal Mechanism

The SIM operates through a structured, multi-stage sequence linking pre-institutional characteristics to eventual persistence:

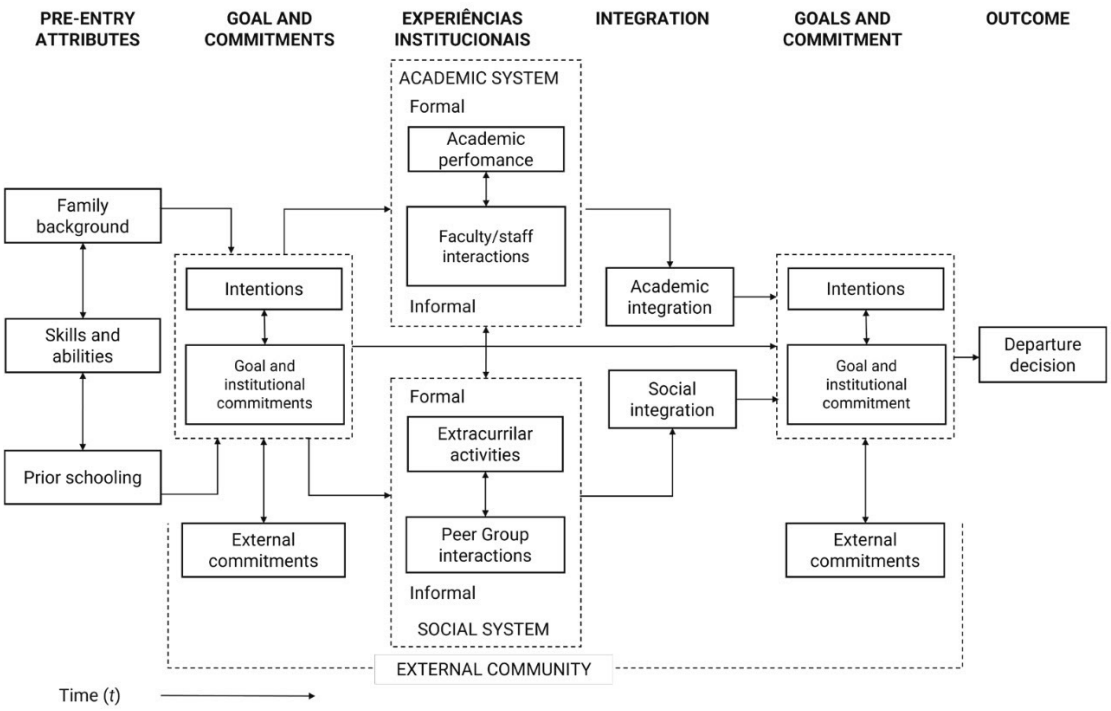

Figure 2.1: Tinto's Model of Institutional Departure

1. **Pre-Entry Attributes:** Students enter the institution carrying three foundational background variables: family background, individual skills and abilities, and prior schooling. These attributes do not directly determine persistence but collectively shape the intellectual and social disposition the student brings to the institution (Tinto, 1975).

2. **Initial Goals and Commitments:** Pre-entry attributes feed into two parallel commitment structures. *Intentions and goal commitments* reflect the student's desire to earn a degree, while *goal and institutional commitments* reflect the de-

sire to graduate from a specific institution. Alongside these sits a third force—*external commitments*—representing obligations outside the institution, such as family or employment, which compete with academic integration from the outset (Tinto, 1975, 1993a).

3. **Institutional Experiences:** Once enrolled, students interact simultaneously with two systems. The *Academic System* operates through both formal channels—academic performance—and informal channels—faculty and staff interactions. The *Social System* operates through formal extracurricular activities and informal peer group interactions. Both systems are embedded within and influenced by the broader *External Community* (Tinto, 1993b).

4. **Integration:** The quality of institutional experiences determines the degree of integration achieved along two dimensions. *Academic integration* results from engagement with the Academic System, while *social integration* results from engagement with the Social System. These two dimensions are interdependent—weakness in either increases the probability of departure (Nicoletti, 2019; Tinto, 1975).

5. **Re-evaluated Goals and Commitments:** Integration feeds back into a revised set of commitments—updated intentions, goal and institutional commitments, and external commitments—which are now reconfigured by the student's actual lived experience within the institution. This re-evaluation stage is what makes the model longitudinal and iterative rather than static: commitments are not fixed at entry but continuously reshaped over time (Nicoletti, 2019; Tinto, 1993b).

6. **Departure Decision (Outcome):** The re-evaluated commitments produce the final outcome: a *departure decision*. If the cumulative weight of academic and social integration has reinforced the student's goal and institutional commitment, persistence follows. If integration has been insufficient and external commitments or personal costs outweigh the perceived benefits of continued enrollment, departure follows. Critically, the entire process unfolds across time (t), underscoring

that dropout is a process, not a moment (Tinto, 1975).

2.3.1.2 The Two Pillars of Integration

Tinto distinguishes two primary and interdependent dimensions of integration, both of which must be achieved to sustain persistence (Tinto, 1993b):

1. **Academic Integration:** encompasses both structural and normative dimensions. Structural integration is measured by grade performance (GPA) relative to institutional standards, while normative integration reflects intellectual development and alignment with the academic community's values. Grades function not merely as performance records but as institutional signals of belonging within the academic system; a declining GPA is therefore not simply a performance metric but a symptom of deteriorating academic integration (Bento Júnior et al., 2025; Tinto, 1975).

2. **Social Integration:** reflects the quality of peer-group interactions and faculty relationships. Research on blended and online learning environments has confirmed that when instructors provided timely support, appropriate feedback, and interactive engagement, students were substantially more likely to persist (Ivankova & Stick, 2007). Furthermore, studies measuring LMS engagement have established that completers consistently demonstrate significantly higher levels of participation in learning activities—measured through discussion posts, resource access, and login frequency—than withdrawers, validating these behavioral signals as digital proxies for integration (Morris et al., 2005b).

Limitation Critical Assessment

Despite its canonical status, the SIM carries limitations directly relevant to this study. Its key constructs—integration, commitment, and fit—lack precise operational definitions, creating interpretive challenges when translating the model into computable variables (Nicoletti, 2019). The model was also developed within a Western, residential university context and requires careful recalibration before application at Strathmore

University.

More fundamentally, traditional SIM-based research relies on retrospective, correlational designs: the model identifies what predicts departure but does not generate prospective, real-time predictions for individual students. This retrospective limitation is the precise gap that machine learning addresses— where the SIM identifies the structural risk factors, ML determines when those factors are currently activating for a specific student.

2.3.1.3 Operationalization in This Study

This study operationalizes Tinto's model within Strathmore's digital infrastructure by systematically mapping each theoretical construct onto a measurable institutional variable. The mapping is deliberate and bidirectional: every predictor variable selected for the ML models corresponds to a specific SIM construct, ensuring that algorithmic outputs retain theoretical interpretability.

2.3.2 Astin's Input-Environment-Output (I-E-O) Model

Alongside Tinto's SIM, Alexander Astin's Input-Environment-Output (I-E-O) model provides a complementary and practically powerful theoretical lens for this study. Originally developed as a conceptual guide for institutional assessment, the model was empirically consolidated in Astin's landmark study *What Matters in College? Four Critical Years Revisited*, which drew on surveys of more than 20,000 students, 25,000 faculty members, and 200 institutions to examine how academic programmes, faculty, peer groups, and institutional policies shape student development and outcomes (Astin, 1993). The model's philosophical foundation is a *talent development* approach to institutional excellence: quality is not measured by institutional prestige or resource endowment, but by the degree to which the institution succeeds in developing the potential of the students it admits (Astin & Antonio, 2012). This orientation makes the I-E-O model especially relevant to the Strathmore context, where the goal is not to rank students but to identify and support those whose potential is at risk of being lost.

2.3.2.1 Structure of the Model

The I-E-O model organizes the variables relevant to student success into three sequential, interacting components (Astin & Antonio, 2012):

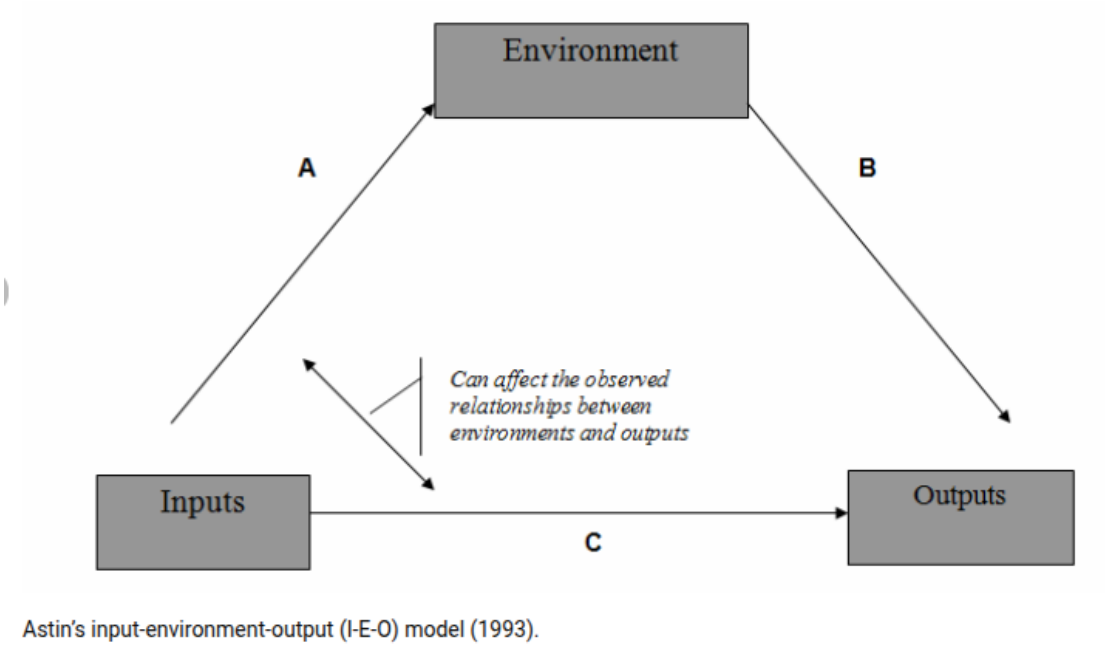

Figure 2.2: Astin's Input-Environment-Output (I-E-O) Model (1993)

1. **Inputs** refer to the personal qualities and characteristics students bring with them at the point of entry. This includes demographic attributes (age, sex), family background, prior academic achievement such as secondary school GPA, and initial motivations and goals. Astin's framework treats no single input as deterministic; rather, inputs establish the baseline against which the effects of institutional experiences must be measured (Astin & Antonio, 2012).

2. **Environment** encompasses the full range of institutional experiences, programmes, policies, and interactions to which the student is exposed during enrollment. This includes both tangible elements (classrooms, library resources, residential halls) and intangible elements (teaching pedagogy, administrative policies, peer group composition, and the quality of faculty interaction). Environmental variables are particularly significant in the I-E-O framework because they represent the factors that educators and institutional administrators can directly control to maximize

talent development (Astin & Antonio, 2012). Applied research using this framework has confirmed that student-advisor interaction, student-faculty interaction, extracurricular participation, and library utilization are each positively and independently associated with first-to-second year retention (Xiao et al., 2020).

3. **Outputs** (or Outcomes) represent the measurable results of the educational programme: academic achievement, retention, degree completion, and the full range of skills, values, and attitudes the institution aims to develop. In the I-E-O framework, outcomes are understood as the product of how environmental conditions transform a student's initial input characteristics. Evaluating outcomes without accounting for inputs is explicitly identified by Astin as an incomplete assessment design, as it fails to capture how much students actually developed relative to their starting point (Astin & Antonio, 2012).

2.3.2.2 Methodological Logic: The Natural Experiment

A defining methodological feature of the I-E-O model is its use of what Astin terms *natural experiments*. Because students cannot be randomly assigned to different institutional environments, the model relies on complex multivariate statistical analyses—such as multiple logistic regression—to control for input differences and isolate the true effect of environmental variables on outcomes (Astin & Antonio, 2012). This approach offers two practical advantages. First, it avoids the artificial conditions of laboratory experiments by studying real institutional variation. Second, it allows researchers to examine the simultaneous effects of many environmental variables, which is precisely the analytical challenge confronted in this study's multi-stream data pipeline.

2.3.2.3 Complementarity with Tinto's Model

The I-E-O model and Tinto's SIM are theoretically complementary rather than competing. Tinto explains *why* students depart through the mechanism of diminishing integration; Astin provides the *variable architecture*— the structured categorization of inputs, environmental conditions, and outcomes—that a predictive model must measure (Bento

Júnior et al., 2025; Nicoletti, 2019). This complementarity is particularly significant for the three-outcome prediction framework of this study. Dropout risk maps most directly to Tinto's integration failure mechanism. Course failure risk and programme delay risk, however, are better understood through Astin's environmental lens: they reflect whether the institution's academic policies, attendance requirements, and support resources are functioning as effective environmental buffers against disengagement.

2.3.2.4 Operationalization in this Study

The I-E-O model maps onto the three institutional data streams in this study's analytics pipeline:

1. **Inputs** are operationalized through Academic Management System (AMS) variables: entry qualifications, prior GPA trajectory, school of enrollment, and year of study—establishing baseline characteristics against which environmental effects are assessed (Astin & Antonio, 2012).

2. **Environment** is operationalized through LMS engagement metrics and attendance records. Strathmore's 67% attendance threshold and 2.0 GPA floor function as measurable environmental constraints explicitly reflected in the predictive models. Prior research confirms that engagement variables are positively associated with retention (Xiao et al., 2020).

3. **Outputs** are the three predicted risk outcomes: dropout, course failure, and programme delay—extending Astin's framework beyond the graduation/dropout binary to capture the fuller spectrum of academic difficulty (Astin & Antonio, 2012).

2.3.2.5 Limitation Critical Assessment

The I-E-O model carries documented limitations relevant to this study. Developed within a Western, residential, four-year university context, it assumes full-time on-campus participation as the primary environmental channel (Astin, 1993)—an assump-

tion requiring recalibration for commuter, part-time, or non-Western settings (Karp et al., 2008).

The model's linear input-to-environment-to-output structure further oversimplifies student development. In practice, outputs such as academic performance continuously reshape subsequent engagement, creating feedback loops a static framework cannot capture (Astin & Antonio, 2012). These limitations do not invalidate the I-E-O model as a theoretical anchor; rather, they motivate a longitudinal machine learning architecture that continuously re-evaluates risk as new engagement and performance data arrives—compensating for the model's structural simplifications in live institutional deployment.

2.4 Machine Learning in Educational Data Mining (EDM)

2.4.1 Overview of EDM

Educational Data Mining (EDM) is a rapidly developing interdisciplinary field that investigates techniques for identifying significant patterns in large-scale educational datasets to enhance teaching, learning, and institutional efficacy (Bello et al., 2025). Within this domain, Machine Learning (ML) has become a foundational tool, shifting EDM from theoretical investigation into a practical instrument for data-driven decision-making. By applying computational analysis to data originating in educational contexts—such as student information systems, Learning Management Systems (LMS), and behavioral interaction logs—ML algorithms identify hidden patterns to forecast future student performance and behavior (Dirin & Saballe, 2022b). A recent systematic review of 72 peer-reviewed studies concluded that research in this domain has matured significantly, with ensemble methods, deep learning, and explainable AI now representing the three dominant technical paradigms (**Aldowah2026**).

2.4.2 Core Applications of Machine Learning in Higher Education

The primary applications of machine learning in EDM converge on four objectives that are directly relevant to this study:

2.4.2.1 Performance Prediction

Academic performance prediction involves forecasting outcomes—such as final course grades, semester GPA, or examination results—by analyzing historical academic records, engagement trajectories, and behavioral patterns. These models allow institutions to move beyond end-of-term assessment toward continuous, data-informed monitoring of student progress (Kwakye, 2025).

2.4.2.2 Risk Identification

Early risk identification applies ML models to flag at-risk students early in a term, enabling timely and targeted interventions before failure becomes irreversible. This is the primary objective of the Early Warning System (EWS) architecture developed in this study (Bello et al., 2025).

2.4.2.3 Study Path and Programme Guidance

Programme guidance uses algorithmic predictions of student preferences, strengths, and motivational orientations to recommend appropriate academic specializations. Research at Haaga-Helia University demonstrated that Random Forest (94% accuracy) and Decision Tree (93% accuracy) are effective classifiers for study path selection, outperforming logistic regression when non-linear motivational and demographic features are included (Dirin & Saballe, 2022b).

2.4.2.4 Adaptive and Personalized Learning

Personalized learning supports the creation of intelligent environments that tailor content, pacing, and feedback to individual student profiles based on continuous engagement monitoring (Kwakye, 2025).

2.4.3 Machine Learning Algorithms and Techniques

2.4.3.1 Machine Learning Algorithms

Machine Learning especially its type of Supervised learning, is the most widely adopted paradigm in EDM, primarily applied to classification tasks—predicting categorical outcomes such as pass/fail, dropout/persist, or risk level—and regression tasks that produce continuous scores (Bello et al., 2025; Dirin & Saballe, 2022b). Three families of supervised learning algorithms are prominent in the EDM literature: ensemble methods, deep learning models, and traditional classifiers.

1. Ensemble Methods such as Random Forest, XGBoost, and Gradient Boosting are consistently identified as top performers in EDM contexts due to their ability to manage complex, non-linear relationships in educational data more effectively than baseline linear models. Bello et al. (2025) demonstrated this directly: on a dataset of 1,163 student records, a baseline Linear Regression model achieved an R^2 of only .32, while Random Forest achieved .69 and XGBoost achieved .70, confirming the superior generalisation capacity of ensemble methods (Bello et al., 2025).

2. Deep Learning models—including Multi-Layer Perceptrons (MLP), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks—are increasingly utilised to capture intricate temporal interactions between features. While these architectures often achieve high predictive accuracy, they operate as "black boxes," providing limited insight into the reasoning behind individual predictions, which significantly limits their institutional adoption in advisory contexts (**Aldowah2026**; Kwakye, 2025).

3. Traditional Classifiers including Decision Trees, Support Vector Machines (SVM), Naïve Bayes, and Logistic Regression remain widely used for their relative simplicity, computational efficiency, and inherent interpretability. Their performance, however, is generally outpaced by ensemble methods when the underlying data

contains non-linear feature interactions (Bello et al., 2025).

2.4.3.2 Techniques

Selecting the right algorithm is necessary but insufficient; the techniques governing how that algorithm is trained and validated determine whether its performance generalises reliably to unseen students. Of the three algorithm families reviewed above, ensemble methods in particular require careful deployment discipline—their capacity to fit complex non-linear patterns makes them equally capable of overfitting when training and evaluation procedures are not rigorously controlled. Standard practice in EDM therefore applies four core techniques to ensure that reported performance reflects genuine generalisation rather than dataset-specific memorisation (**Aldowah2026**; Bello et al., 2025).

1. **Stratified train/test splitting** preserves the empirical class ratio across training and holdout partitions, preventing artificially optimistic evaluation on imbalanced datasets.

2. k**-fold cross-validation** assesses generalisation across multiple data subsets rather than a single holdout, producing more stable performance estimates.

3. **SMOTE** (Synthetic Minority Oversampling Technique) addresses the class imbalance characteristic of student risk datasets—where at-risk students constitute a minority—by generating synthetic minority-class instances during training.

4. **Hyperparameter tuning** optimises each algorithm's configuration prior to final evaluation, ensuring that reported performance reflects a properly calibrated model rather than default settings (**Aldowah2026**; Bello et al., 2025).

2.4.4 Data Sources and Feature Engineering

2.4.4.1 Data Sources

A data source is any institutional system that generates raw student records—such as grade histories, login activity, or attendance logs—that can be extracted for predictive modelling. Without reliable and comprehensive data sources, no algorithm or engineering procedure can produce valid risk predictions. ML models in EDM rely on three primary institutional data sources: Academic Management Systems, which supply enrollment records, GPA trajectories, and academic standing data; Learning Management Systems, which contribute behavioral engagement signals including login frequency, resource access, and assignment submission patterns; and clickstream logs from digital platforms, which capture fine-grained temporal interaction data (Kwakye, 2025). Together, these sources provide the longitudinal, multi-dimensional student record that predictive models require.

2.4.4.2 Feature Engineering

Raw institutional records cannot be fed directly into predictive models; they must first be transformed into structured, model-ready input variables that meaningfully represent student behaviour and academic trajectory. This transformation process is **feature engineering**, and it constitutes the critical intermediate step between data extraction and model training. Typical engineered features include attendance rates, prior grade trajectories, assignment submission timeliness, and temporal engagement patterns.

Research has established that *time-based features*—such as the intervals between LMS interactions, the recency of last login, and the "burstiness" of activity relative to assignment deadlines—carry significant predictive power, often capturing early disengagement signals that static features such as a single GPA value cannot detect (Dirin & Saballe, 2022b). To ensure model stability and address the class imbalance that characterises most student risk datasets—where at-risk students constitute a minority—standard pre-processing techniques including normalisation, one-hot encoding, and the

Synthetic Minority Oversampling Technique (SMOTE) are routinely applied (**Aldowah2026**; Bello et al., 2025).

2.4.5 Explainability, Fairness, and Ethical Considerations

Three foundational concepts structure this subsection. **Explainability** refers to the capacity of a predictive model to provide human-interpretable reasons for its outputs—enabling advisors and administrators to understand, interrogate, and defend risk classifications. Fairness refers to the requirement that model predictions do not systematically disadvantage students on the basis of protected or socioeconomic characteristics. Ethical considerations encompass the broader governance obligations—including data privacy, accountability, and human oversight—that arise when algorithmic outputs are used to make or inform decisions affecting students' academic futures. As ML becomes embedded in institutional decision-making, all three have emerged as critical prerequisites for responsible deployment (Bello et al., 2025; Kwakye, 2025).

2.4.5.1 Explainability and the Black Box Problem

Complex ensemble and deep learning models produce predictions without human-interpretable reasoning, creating an institutional trust deficit that prevents adoption by advisors and administrators who cannot interrogate or defend the model's outputs (Kwakye, 2025). To address this, researchers now employ XAI (Explainable AI) techniques—most prominently SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and Partial Dependence Plots (PDP)—to provide feature-level explanations for individual predictions (Kwakye, 2025). A Springer study applying CatBoost–SHAP to student academic crisis warning demonstrated that explainable predictions provide intuitive decision support and guidance assistance for education administrators, validating SHAP's role as a trust-enabling layer in institutional deployments (Zhang & Tao, 2021).

2.4.5.2 Fairness and Bias

Predictive models trained on historically biased institutional data risk reinforcing systemic inequalities, particularly for students from disadvantaged socioeconomic backgrounds. Without deliberate fairness auditing, a model may achieve high aggregate accuracy while systematically misclassifying vulnerable subpopulations—a failure mode that is especially consequential when model outputs determine which students receive scarce advising and support resources (**Bello20bello202525**).

2.4.5.3 Ethical Considerations

Two interrelated ethical obligations govern institutional deployment of student analytics systems. First, human-in-the-loop deployment requires that ML models function as decision-support tools that complement rather than replace human pedagogical judgment (Kwakye, 2025). The advisor remains the responsible agent; the model supplies evidence. This principle directly informs the role-based dashboard architecture of this study, where SHAP outputs are presented as advisor-facing explanations rather than autonomous risk verdicts. Second, data privacy requires compliance with data protection frameworks—including GDPR in European contexts and equivalent national regulations—mandating robust data anonymisation, access controls, and transparent data governance policies (Kwakye, 2025).

2.4.6 Existing ML-based predictive analytics for at-risk students

2.4.6.1 Georgia State University: GPS Advising System

Institutional Context and Motivation

Georgia State University (GSU) is a large public research university in Atlanta (Georgia State University, 2024b). It enrolls more than 50,000 students—of whom 58% qualify as low-income by federal standards, 60% are non-white, and 30% are first-generation college students—provides the most extensively documented and empirically evaluated implementation of ML-based predictive analytics for student retention in higher edu-

cation (Georgia State University, 2024b; Renick, 2017). Prior to 2012, GSU's annual dropout rate stood at approximately 5,600 students per year, with adviser-to-student ratios as high as 1:1,000. Students—particularly those from minority and low-income backgrounds—were failing and withdrawing without ever seeking help, because the institution had no mechanism for identifying them before failure had already occurred (Kamenetz, 2016).

System Architecture and Methodology

In 2011, GSU entered a partnership with the Education Advisory Board (EAB)—a specialist higher education analytics consultancy—to develop what would become the GPS (Graduation and Progression Success) Advising system (Kamenetz, 2016; Wan, 2019b). The methodology proceeded in three phases:

Phase 1 — Historical Data Analysis: EAB analysts processed ten years of GSU institutional records, encompassing 2.5 million course grades and 140,000 student records, to identify patterns statistically associated with student attrition and programme delay (Association of Governing Boards, 2019; Kamenetz, 2016).

Phase 2 — Risk Factor Development: From this historical analysis, a library of more than 800 risk markers was developed. These markers span a wide range of behavioral and academic signals: course registration anomalies (such as a senior enrolling in a course that does not count toward their degree), grade patterns, financial account status, LMS engagement changes, and attendance indicators derived from Wi-Fi and LMS login monitoring (EdTech Magazine, 2019; Georgia State University, 2024b). Risk markers range from *repeated historical patterns*—behaviours that have historically preceded dropout across thousands of prior students—to *specific points of change*, such as a sudden drop in grades or a significant reduction in LMS usage (Georgia State University, 2024b).

Phase 3 — Operational Deployment: The GPS system went live in August 2012. It operates as a nightly batch process: student grades, records, and behavioral indicators are updated every night, and the system generates push-notification alerts to the

assigned academic adviser whenever a student triggers one or more risk markers (Association of Governing Boards, 2019; Kamenetz, 2016). Advisers are expected to initiate contact with the flagged student within 48 hours (EdTech Magazine, 2019). The system is delivered through EAB's proprietary web platform, *Navigate*, which presents each student's risk score on a color-coded dashboard (green/yellow/red) alongside the specific alert triggers (The Signal, 2019).

Outcomes and Impact

The results of GPS Advising are among the most cited in the student success literature. Since its launch, GSU has recorded the following verified outcomes (Association of Governing Boards, 2019; Georgia State University, 2024b; National Institute for Student Success, 2024):

- Graduation rates increased by 23 percentage points over the decade following implementation, with four-year graduation rates specifically improving by 7.8 percentage points.
- The system generates more than 90,000 adviser-initiated interventions per academic year, compared to approximately 1,000 annual adviser meetings before 2012.
- Over 250,000 one-on-one meetings between advisers and students have been prompted by GPS alerts since 2012.
- Achievement gaps based on race, ethnicity, and income level have been eliminated: Black, Hispanic, first-generation, and low-income students now graduate at rates at or above the overall student body rate—the only national university in the United States to have achieved this outcome.
- A Boston Consulting Group analysis confirmed that the programme generates more revenue than it costs, with each one-percentage-point increase in retention generating approximately $3.18 million in additional tuition revenue annually (Wan, 2019b).

Limitations and Critical Assessment

Proprietary and Non-Transferable Architecture: The most consequential limitation is that GPS Advising is built on EAB's proprietary *Navigate* platform, for which GSU paid a heavily discounted rate of $150,000 per year as an early adopter (Wan, 2019b). Market-rate licensing costs are substantially higher for most institutions in the Global South, including Kenyan universities operating under severe resource constraints. This creates a fundamental equity problem: the institutions most in need of student success technology are precisely those least able to afford commercial vendor solutions.

Lack of Model Transparency: Because the underlying algorithmic logic of the *Navigate* platform is proprietary, institutions cannot inspect, audit, or modify the predictive models. This opacity prevents the kind of institution-specific calibration that this study argues is essential: Strathmore University's 67% attendance threshold and 2.0 GPA floor require that predictive models be explicitly tuned to local policy parameters, which is impossible within a black-box commercial system (The Signal, 2019; Wan, 2019b).

Resource-Intensive Implementation: GSU's success was not solely the product of technology. More than 90% of the total programme cost was in staffing: GSU hired 42 new academic advisers in the first six months, reducing the adviser-to-student ratio from 1:1,000 to approximately 1:300 (Kamenetz, 2016). This level of staffing investment is not replicable in resource-constrained contexts, and it raises the question of whether the technology or the additional human capacity was the primary driver of improved outcomes.

Relevance to This Study

The Georgia State case establishes two foundational propositions for this research. First, it provides proof of concept at institutional scale: predictive analytics, when properly implemented and supported by human advisory capacity, can dramatically improve retention and eliminate equity gaps (Georgia State University, 2024b). Second, and equally importantly, its limitations define the design requirements for the current study.

The Strathmore tool intended to be developed in this thesis is therefore not simply inspired by the Georgia State model—it is explicitly designed to resolve the transferability and transparency gaps that prevent that model from being deployed in a Kenyan institutional context.

2.4.6.2 Morocco: A Modular and Explainable Machine Learning Pipeline at Mohammed V University

Institutional Context and Motivation

While the Georgia State GPS Advising system demonstrates the institutional impact achievable through predictive analytics at scale, its proprietary architecture and resource requirements limit its transferability to institutions in the Global South. A critically important and more recent contribution to the EDM literature—directly relevant to the African higher education context—is a modular dropout prediction pipeline developed at the Mohammadia School of Engineers, Mohammed V University in Rabat, Morocco, by Bettahi, Belouadha, and Harroud (2025). This study is geographically proximate to the Kenyan context, uses open-source tools, and explicitly addresses the governance and explainability requirements that the GSU system does not (Bettahi et al., 2025).

Dataset and Institutional Setting

The study used a longitudinal institutional dataset of 17,883 undergraduate records drawn from multiple cohorts at a Moroccan HEI. After cleaning and feature curation, the modeling matrix comprised 91 features integrating four categories of student data: demographic and programme descriptors; academic progression variables (cumulative GPA, credits earned, course failures, and passed course counts); engagement and attendance metrics; and financial and administrative signals encoded as binary hold flags (`HAS_*` variables) with paired timestamp durations. The positive class—students recorded by the registrar as having discontinued their studies—comprised 2,856 students (16.0%), producing a substantially imbalanced classification problem typical of real institutional data (Bettahi et al., 2025).

Methodology and Pipeline Architecture

The pipeline was implemented entirely in Python 3.12.0 using open-source libraries, specifically scikit-learn 1.7.1, xgboost 3.0.3, and shap 0.48.0. This is a significant methodological distinction from the proprietary EAB *Navigate* platform used by Georgia State: every component of the Moroccan pipeline is reproducible, auditable, and transferable without licensing costs (Bettahi et al., 2025).

The six-stage modular architecture proceeded as follows (Bettahi et al., 2025):

1. **Data Ingestion and Cleaning:** Automatic delimiter detection; removal of columns with $> 90\%$ missingness; median imputation for numeric fields; label encoding for categorical variables with an explicit <MISSING> token; date normalization to elapsed days; and leakage prevention by dropping direct student identifiers.

2. **Feature Engineering:** Transformation of raw institutional records into 91 structured predictors, including temporal features such as hold durations and derived academic momentum indicators.

3. **Train/Test Split:** A stratified 80/20 split yielding 14,306 training and 3,577 test students, preserving the empirical class ratio in both partitions.

4. **Model Benchmarking:** Nine algorithms were evaluated: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), k-Nearest Neighbors (k-NN), Support Vector Machine (SVM), Gradient Boosting (GB), XGBoost, Naïve Bayes (NB), and Multilayer Perceptron (MLP).

5. **Explainability Layer (SHAP):** SHAP TreeExplainer applied to all tree-based models on the full 3,577-test-case set; KernelExplainer with k-means background applied to MLP on a 300-sample subset.

6. **Reporting and Visualization:** Structured output directories storing confusion matrices, ROC curves, SHAP plots, performance summaries, and reproducibility logs.

Model validation used stratified 5-fold cross-validation with a fixed random seed (42). A single operating threshold τ was selected per model by maximizing F1 on cross-validation folds and then fixed for all test and subgroup reporting (Bettahi et al., 2025).

Results and Performance

Table 2.2 summarizes the headline test-set performance of all nine models.

Table 2.2: Model Performance on the Moroccan HEI Test Set (Bettahi et al., 2025)

Model	Accuracy	Precision	Recall	F1	AUC-ROC
XGBoost	0.970	0.879	**0.944**	**0.911**	**0.993**
Random Forest	0.968	0.919	0.877	0.898	0.989
MLP	0.964	0.891	0.884	0.888	0.979
Gradient Boosting	0.963	0.909	0.855	0.881	0.988
Decision Tree	0.961	0.862	0.900	0.881	0.936
SVM	0.954	0.816	0.921	0.865	0.983
k-NN	0.948	0.892	0.767	0.825	0.947
Logistic Regression	0.917	0.682	0.904	0.777	0.968
Naïve Bayes	0.689	0.322	0.858	0.468	0.881

XGBoost achieved the highest performance across all primary metrics: AUC-ROC of 0.993, F1-score of 0.911, and recall of 0.944. McNemar's paired significance testing confirmed that XGBoost was statistically superior to Gradient Boosting ($p = 0.008$) on the test set, while the difference from Random Forest was not statistically significant ($p = 0.428$) (Bettahi et al., 2025). To assess generalizability, the full pipeline was replicated on the public UCI Portugal dataset (Realinho et al., 2021), where XGBoost remained top-ranked (F1-score = 0.792, AUC-ROC = 0.922), confirming model ordering preservation across feature-shifted datasets.

Limitation Critical Assessment

The Bettahi et al. pipeline is the most methodologically aligned existing system in the

literature relative to the platform developed in this thesis. It validates XGBoost and SHAP as the dominant tools for structured, imbalanced HEI dropout prediction using entirely open-source infrastructure—directly addressing the transferability limitation of the Georgia State model (Bettahi et al., 2025). The use of Python with open-source libraries (`scikit-learn`, `xgboost`, `shap`) establishes the technical feasibility of deploying a high-performance, explainable EWS without commercial licensing, while the SHAP-based risk versus protective factor distinction provides the explainability architecture template for the role-based advisor dashboards developed in Chapter Four.

However, the study carries two significant limitations that this thesis directly addresses. First, the institution-specific features central to the pipeline—particularly administrative timestamp variables such as `AP_START_DTE` and `AP_END_DTE`—are Moroccan HEI-specific and carry no equivalent semantics in other institutional contexts, limiting the direct transferability of the feature engineering strategy to universities with different administrative infrastructures (Bettahi et al., 2025). This motivates this study's institution-specific feature engineering, which encodes Strathmore's 67% attendance threshold and 2.0 GPA floor as policy-calibrated variables rather than generic proxies. Second, the pipeline predicts a single binary dropout outcome only, failing to capture the graduated spectrum of academic difficulty that students actually experience. This study departs from that binary framing through the simultaneous prediction of three distinct risk outcomes—dropout risk, course failure risk, and programme delay risk—with all models explicitly calibrated to Strathmore University's specific policy thresholds.

2.4.6.3 South Africa: Machine Learning for Engineering Student Success at the University of Johannesburg

One of the earliest investigations of ML-based student performance prediction in the South African context was conducted by Taodzera, Twala, and Carroll at the University of Johannesburg's Department of Electrical and Electronics Engineering Science (Taodzera et al., 2016).

Dataset and Features

The study worked with 1,366 engineering student records drawn from a South African university. Available features spanned demographic and academic entry indicators: high school Mathematics score, Physics score, English score, Life Orientation score, ethnicity, home language, age, school province, and religion. The response variable was defined as the number of subjects passed to qualify for progression to second year—a binary pass/fail classification motivated by the institution's own progression regulations (Taodzera et al., 2016).

Feature Selection

Correlation analysis identified Physics score (r = 0.2391) and Mathematics score (r = 0.1836) as the strongest predictors of first-year success, followed by age (r = 0.1191) and ethnicity (r = 0.0808). The significance of ethnicity as a predictor directly reflects structural inequalities in South African secondary schooling, where historically advantaged schools continue to produce students substantially better prepared for the quantitative demands of engineering study (Taodzera et al., 2016). Wrapper-based feature selection confirmed Mathematics and Physics as the primary predictors, with ethnicity and school province also identified as significant.

Models and Results

Five algorithms were evaluated: Logistic Regression (LR), Naïve Bayes (NB), Artificial Neural Network (ANN), Support Vector Machine (SVM), and Decision Tree (DT). The DT algorithm performed best with 65.86% accuracy, an F1-score of 0.658, and a Kappa statistic of 0.316—classifying it as a fair predictor. SVM ranked second (64.63%), followed by NB (63.41%) and LR (63.17%). ANN performed weakest (60.24%) (Taodzera et al., 2016).

Limitation Critical Assessment

The Taodzera et al. study makes two contributions of direct relevance to this thesis. First, it demonstrates that standard ML algorithms can be operationalized in a resource-limited African institutional setting using only enrollment-time features available at the

point of admission—without requiring LMS integration, real-time data pipelines, or proprietary vendor platforms. Second, the identification of school province and ethnicity as statistically significant predictors highlights a structural feature of African educational data that does not appear in the North American literature: that secondary schooling inequality is itself a measurable risk factor that encodes into university-level performance data.

However, the absence of SHAP-based or any form of explainability analysis means the model cannot be operationalized as a policy-actionable early warning system. Furthermore, the modest Kappa values (all below 0.32) suggest that all five models classify at a level only slightly above what chance would produce (Taodzera et al., 2016).

2.4.6.4 South Africa: Bayesian Network Classification for Computer Science Student Risk Prediction at the University of Cape Town

At the University of Cape Town's Department of Computer Science, Hassen conducted a study applying Bayesian Network classifiers to predict which second-year Computer Science students that were at risk of failing CSC2001F (Hassen, 2020). This was a core data structures course widely regarded as the critical attrition point in the UCT Computer Science major.

Dataset and Features

The study drew on 728 student records spanning six cohorts from 2013 to 2019. Features were selected through iterative consultation with the course convenor and spanned both pre-entry and early in-course indicators: Matric Mathematics mark, Matric English mark, National Benchmark Test (NBT) scores in Mathematics, Quantitative Literacy and Academic Literacy, first-year Computer Science course final marks (CSC1015F and CSC1016S), the first assignment mark for CSC2001F, and the CSC2001F Test 1 mark (Hassen, 2020).

Methodological Design

Three Bayesian Network models were developed using the Netica platform. The first

was manually constructed from expert knowledge gathered in faculty consultations. The second was generated automatically using Netica's Tree Augmented Naïve Bayes (TAN) structure learning algorithm. The third was a combined model synthesising insights from both. A notable methodological contribution was the adoption of incremental training, wherein each cohort's model was trained exclusively on all preceding years' data and evaluated on the subsequent cohort — replicating the temporal constraints of genuine deployment (Hassen, 2020).

Models and Results

The combined model produced the most instructive outcome: while its accuracy (83.33%) was marginally below the TAN model's, it achieved a perfect recall score of 1.0 — correctly identifying every at-risk student in the test cohort — alongside an F1 score of 0.40 and a precision of 0.25 (Hassen, 2020).

Limitation Critical Assessment

The Hassen study validates incremental cohort-based training as a deployment-realistic evaluation strategy and establishes recall as the theoretically appropriate primary metric in student support contexts (Hassen, 2020). However, the dataset of 728 records is small by ML standards, and the absence of any SHAP-based explainability means the model cannot generate student-level explanations sufficient to guide individualised advising conversations.

2.4.6.5 Ethiopia: Predictive Analytics at Scale with African Institutional Data

Institutional Context and Motivation

A more recent study from the East African context was conducted by Esmael (2024) at Wollo University and the Kombolcha Institute of Technology in Ethiopia (Esmael, 2024). The study was motivated by the growing need to leverage institutional data for improving student outcomes in African higher education, where modern learning institutions face persistent challenges in analysing performance and identifying at-risk students early. The dataset comprised 32,582 student records from academic years

2017–2022, extracted from the institutional student information portal, with features including gender, region, entrance examination results, number of previous attempts, studied credits, disability status, and final result (Esmael, 2024).

Methodology

The study employed four classification algorithms—Support Vector Machine (SVM), Decision Tree, Naïve Bayes, and K-Nearest Neighbors (KNN)—evaluated against one another after hyperparameter tuning via grid search. K-means clustering with Davies' Bouldin method was applied first to identify significant features and cluster students into performance groups. Data preprocessing involved cleaning, categorical encoding, and data reduction, while repeated 10-fold cross-validation was used to reduce sample-related bias. Hyperparameter optimisation was applied to all four classifiers to enhance predictive accuracy beyond baseline configurations (Esmael, 2024).

Models and Results

SVM achieved the highest post-tuning accuracy of 96.0% (precision 0.94, recall 0.98, Kappa 0.94), followed by Decision Tree at 93.4% and KNN at 87.4%. Naïve Bayes produced the lowest accuracy at 83.3%, attributed to its strong independence assumption between features. Hyperparameter tuning improved all models substantially—Decision Tree accuracy rose from 90.9% to 93.4% and Naïve Bayes from 77.3% to 83.3%. Entrance examination results and studied credits emerged as the strongest predictors, consistent with findings from both the South African and Moroccan contexts that prior academic achievement and engagement intensity are the most transferable cross-institutional risk signals (Esmael, 2024).

Limitation and Critical Assessment

The Esmael study makes two contributions of direct relevance to this thesis. First, it confirms that classical ML classifiers can achieve high predictive accuracy on large-scale African institutional data using only demographic and entry-level features, without requiring LMS integration or commercial analytics platforms. Second, it demon-

strates that hyperparameter tuning is a critical technique for improving model performance on educational datasets, directly validating the tuning strategy adopted in this study.

However, the study carries significant limitations. The feature set is restricted entirely to enrollment-time and demographic variables, with no longitudinal academic performance or engagement indicators included—meaning the model cannot be updated as in-course data becomes available, limiting its utility as a dynamic early warning system. More critically, the absence of any explainability layer means that model outputs cannot be translated into student-specific intervention recommendations: an adviser cannot determine which specific factors triggered a risk classification for an individual student (Esmael, 2024).

2.4.6.6 Kenya: Business Intelligence and Decision Tree Classification for Student Retention Prediction — case of Machakos University College

One of the few published East African studies applying machine learning techniques to student retention prediction in a higher education context was conducted by Ngemu, Odongo, Omulo, and Manderick, drawing on data from Machakos University College in Kenya (J. M. Ngemu et al., 2015). The J48 decision tree outperformed all competitors, achieving a cross-validation accuracy of 94.8% and a recall of 94.4%. The dominance of fees payment difficulty as the single most informative predictor — with a gain ratio of 0.424 — positions financial exclusion as the primary mechanism of student attrition in this Kenyan institutional setting (J. M. Ngemu et al., 2015). However, the dataset of 270 instances is extremely small, and the study provides no class imbalance handling, no SHAP-based explainability, and no longitudinal or cohort-based evaluation.

2.4.6.7 Research Gaps Table

Study	Context	Technique Used	Key Predictors	Identified Gaps
Georgia State (Georgia State University, 2024b)	USA, 50,000+	800+ risk markers; EAB Navigate	Grades, LMS, financial	Proprietary; non-transferable; unaffordable
Bettahi et al. (Bettahi et al., 2025)	Morocco, 17,883	XGBoost, RF, SHAP	GPA, credits, hold flags	Single binary outcome only
Taodzera et al. (Taodzera et al., 2016)	S. Africa, 1,366	DT, SVM, ANN	Physics, Maths, age	No longitudinal features; no explainability
Hassen (Hassen, 2020)	S. Africa, 728	Bayesian Network	Assignment 1, Test 1	Small dataset; precision = 0.25; no SHAP
Esmael (Esmael, 2024)	Ethiopia, 32,582	RF, GB, DT	Entrance exam, credits	No explainability; no SHAP or LIME
Ngemu et al. (J. M. Ngemu et al., 2015)	Kenya, 270	J48, NB, SVM (WEKA)	Fee payment, age, health	Tiny dataset; enrollment-time only; no SHAP

Table 2.3: Summary of Research Gaps in Existing ML-Based Student Attrition Studies

2.4.6.8 Relevance to our study Table

Study	Contribution to This Study	Design Element Adopted	Design Element Improved or Departed From
Georgia State (Georgia State University, 2024b)	Proof of concept at scale	Role-based dashboards; risk score updates	Strathmore thresholds; no vendor dependency
Bettahi et al. (Bettahi et al., 2025)	Most methodologically aligned; validates XGBoost + SHAP	Python open-source stack; stratified split	Three outcomes; Strathmore policy features
Taodzera et al. (Taodzera et al., 2016)	ML in African resource-limited setting	Feasibility without proprietary tools	Added LMS features; SHAP explainability
Hassen (Hassen, 2020)	Recall as primary metric; SMOTE	Incremental training; recall priority	Ensemble methods replace Bayesian Networks
Esmael (Esmael, 2024)	Ensemble superiority on African data	RF and GB as primary candidates	SHAP added; LMS engagement included
Ngemu et al. (J. M. Ngemu et al., 2015)	Earliest Kenyan ML study; financial exclusion finding	Tinto grounding; Kenyan deployment	Python pipeline; SMOTE; SHAP; larger dataset

Table 2.4: Relevance of Reviewed Studies to This Study's Design

2.4.6.9 Conceptual Framework

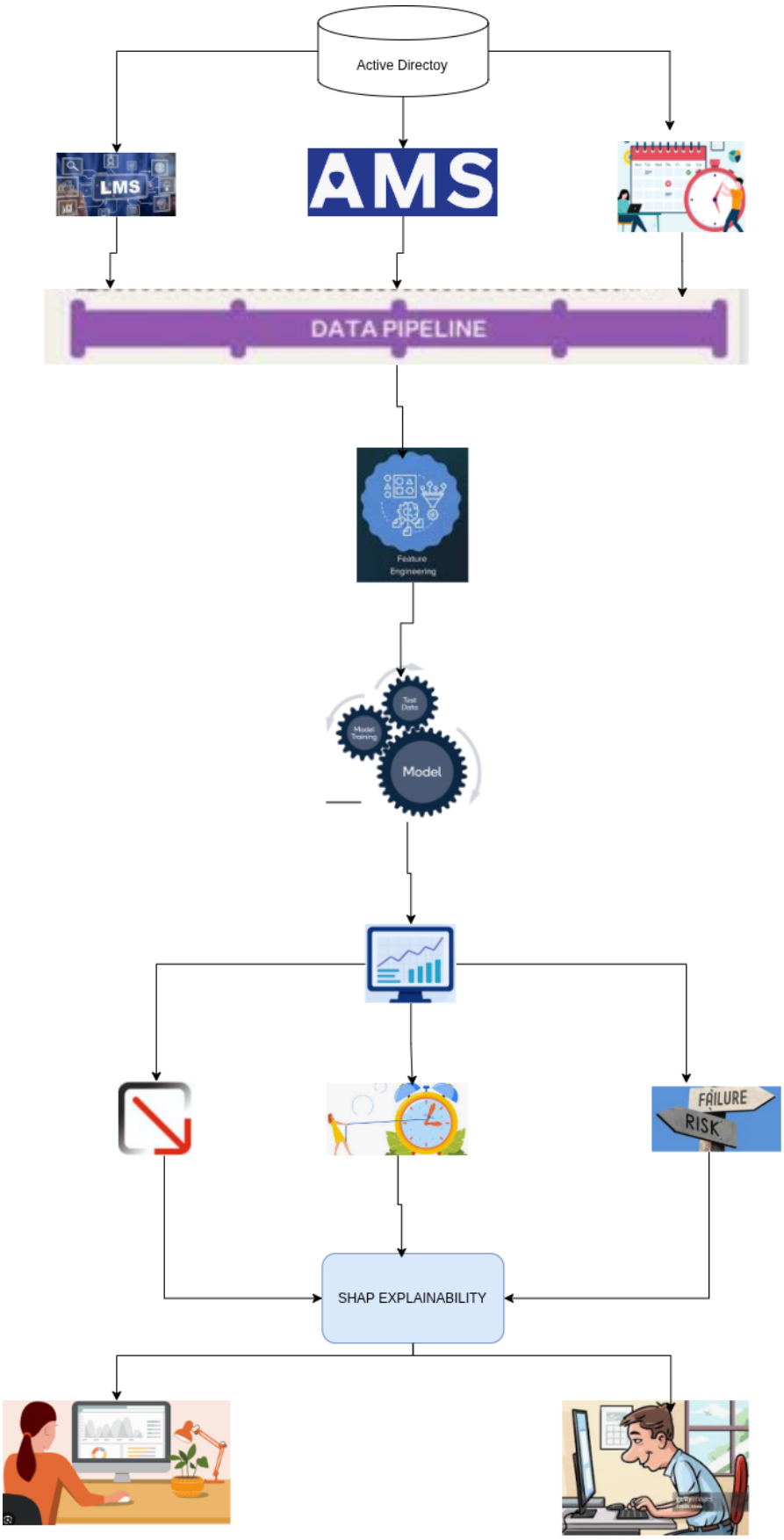

Figure 2.3: Conceptual Framework Diagram

Conceptual Framework Explanation

The solution architecture presented in the above conceptual framework diagram describes an end-to-end pipeline through which raw institutional data is transformed into actionable, explainable risk predictions for Strathmore University's academic support staff. The framework flows strictly from top to bottom, with each stage building on the output of the preceding one.

Stage 1 — Authentication: The pipeline originates from Active Directory, which governs authenticated access to the three institutional source systems, ensuring that only authorised data flows enter the pipeline.

Stage 2 — Data Sources: Three institutional systems operate in parallel: the Academic Management System (AMS), which supplies enrollment records, GPA trajectories, and academic standing data; the Learning Management System (LMS), which contributes behavioral engagement signals including login frequency, resource access, and assignment submission patterns; and the Attendance Tracking Platform, which provides real-time attendance rates and flags students approaching the mandatory 67% examination eligibility threshold.

Stage 3 — Data Pipeline: Raw data extracted from the three source systems is passed through an integrated data pipeline that merges records across systems, resolves inconsistencies, and validates data integrity before any modelling takes place.

Stage 4 — Feature Engineering: The cleaned dataset is transformed into a structured feature matrix. This stage applies normalisation, categorical encoding, SMOTE oversampling to address class imbalance, and the construction of derived indicators such as attendance breach flags, GPA floor proximity scores, and temporal engagement patterns.

Stage 5 — Model Training: Three ensemble algorithms—Random Forest, XGBoost, and Gradient Boosting—are trained on the engineered feature matrix using historical student cohort data from 2021 to 2024. Stratified cross-validation and hyperparameter

tuning are applied to maximise generalisation performance.

Stage 6 — Prediction Engine: The trained models generate three simultaneous risk scores for each currently enrolled student: dropout risk, course failure risk, and programme delay risk. This multi-outcome architecture reflects the graduated spectrum of academic difficulty that students experience, extending beyond the binary dropout classification common in prior literature.

Stage 7 — SHAP Explainability: All three risk scores converge into the SHAP explainability layer, which decomposes each prediction into its contributing feature drivers, identifying both risk factors and protective factors at the individual student level. This transforms opaque model outputs into human-interpretable explanations that advisors can act on.

Stage 8 — Institutional Actors: The pipeline terminates with the human actors who consume the system's outputs through role-based dashboards: the University Administrator, the School Administrator, and the Academic Mentor or Advisor. This human-in-the-loop design ensures that the platform functions as a decision-support tool that augments rather than replaces professional academic judgment.

Taken together, the framework operationalises the transition from Strathmore University's current reactive support paradigm—where students are identified only after breaching academic thresholds—to a proactive, evidence-driven intervention model in which at-risk students are flagged months or weeks before failure occurs, and advisors are equipped with the specific reasons behind each risk classification.

Summary

This chapter established the theoretical and empirical foundations for the study's predictive analytics platform. Global attrition statistics confirmed student dropout as a structurally significant challenge, while Tinto's Student Integration Model and Astin's I-E-O Model provided the conceptual framework mapping institutional data variables to retention outcomes. The review of existing ML-based systems — spanning Geor-

gia State University, Morocco, South Africa, Ethiopia, and Kenya — demonstrated that ensemble methods, particularly XGBoost and Random Forest, consistently outperform classical algorithms, while SHAP-based explainability remains critically underutilized in African contexts. All gaps identified directly inform the platform's design. Chapter Three now presents the research methodology, detailing the data collection strategy, feature engineering process, model development approach, and evaluation framework employed to build and validate the system.

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